

ARIMA

Machine learning

Data → Model

Linear reg
log reg
SVM
DT

Time Series Data → Model

ARIMA
SARIMA, SARIMAX
DL, RNN, Attention, transformer

AR

(Autoregression)

↓

P

[0, 1, 2, 3, ... n]

lag value

PACF (Corrlogram)

I

(Integration)

↓

d

- [0, 1, 2, 3, ... n]

lag value

Differencing

Non Stationary | Stationary

MA

(moving average model)

↓

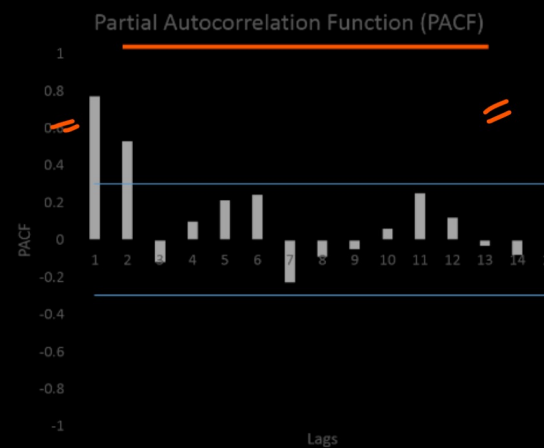
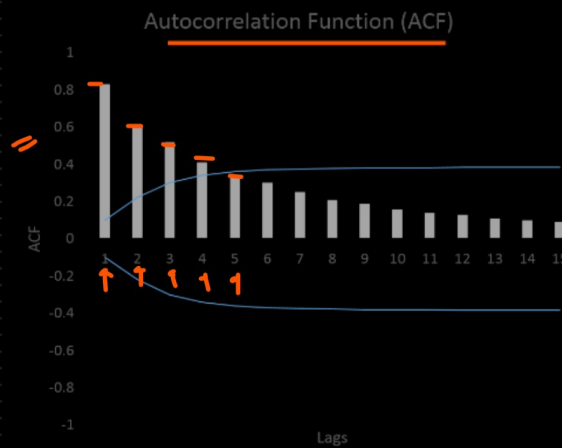
q

[0, 1, 2, 3, ... n]

lag value

ACF (Corrlogram)

$$\left. \begin{aligned} - & \gamma_t, \gamma_{t-1} \\ - & \gamma_t, \gamma_{t-2} \\ - & \gamma_t, \gamma_{t-3} \end{aligned} \right\}$$



$$\hat{=} \frac{\gamma_{t-2} \longleftrightarrow \gamma_t}{\cancel{\gamma_t}}$$

AR (Autoregression)

$$\gamma_t = \psi_1 \gamma_{t-1} + \psi_2 \gamma_{t-2} + \psi_3 \gamma_{t-3} \dots \psi_n \gamma_{t-n} + c$$

AR(1) $\Rightarrow$

$$\gamma_t = \psi \gamma_{t-1} + c$$

lag 1 TS

lag 2

$\rightarrow D_1$	10	NA	NA
D_2	20	$D_1 = 10$	NA
D_3	30	$D_2 = 20$	$D_1 = 10 \Rightarrow$
D_4	40	$D_3 = 30$	$D_2 = 20$
D_5	45	$D_4 = 40$	$D_3 = 30$

$$\begin{aligned} \text{lag} &= 1 \\ \text{AR} &= 1 \\ P &= 1 \end{aligned}$$

20	10
30	20
40	30
45	40
?	45

AR(2) or P=2 or lag 2

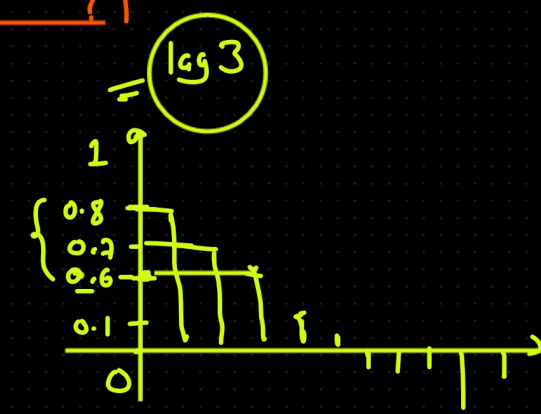
$$\hat{=} \gamma_t = \psi_1 \gamma_{t-1} + \psi_2 \gamma_{t-2} + c$$

$$\Rightarrow \gamma_t = \psi \gamma_{t-1} + c$$

y_t	y_{t-1}	y_{t-2}
30	20	10
40	30	20
45	40	30
(?)	45	40

ACF

AR(3)



D_1	10	lag 2	$y_t - y_{t-2}$
D_2	20		
D_3	30		
D_4	40	lag 1	20
D_5	40		
D_6	60		
			20

$$y_t = y_{t-1}\psi_1 + y_{t-2}\psi_2 + \dots + y_{t-n}\psi_n + c$$

$$y_t = y_{t-1}\psi + y_{t-2}\psi + \dots + y_{t-1c} + c$$

$I \Rightarrow$ Integration $\Rightarrow$ Difference

$D=0$	$D=1$	$D=2$
$y_{t-4} D_1 \leftarrow$		
$y_{t-3} D_2 \leftarrow$	$(D_2 - D_1)$	
$y_{t-2} D_3 \leftarrow$	$(D_3 - D_2)$	
$y_{t-1} D_4 \leftarrow$	$(D_4 - D_3)$	
$y_t D_5 \leftarrow$	$(D_5 - D_4)$	

$$D_1 = (y_t - y_{t-1})$$

$$D_2 = (y_{t-1} - y_{t-2})$$

$$D_2 = (y_t - y_{t-1} - y_{t-2})$$

Moving average $\Rightarrow$
(error)

$$y_t = \varepsilon_t \psi + c$$

$$y_t = \varepsilon_{t-1} \psi_{t-1} + \varepsilon_{t-2} \psi_{t-2} + \varepsilon_{t-3} \psi_{t-3} - - \varepsilon_{t-n} \psi_{t-n} + c$$

ARIMA $\Rightarrow$

$$\hat{y}_t = \underbrace{(y_{t-1} \psi_{t-1} + y_{t-2} \psi_{t-2} + - y_{t-n} \psi_{t-n} + c)}_{\text{AR part}} + \underbrace{(y_{t-1} - y_t - y_{t-2} - - y_{t-n})}_{\text{I(1) part}}$$

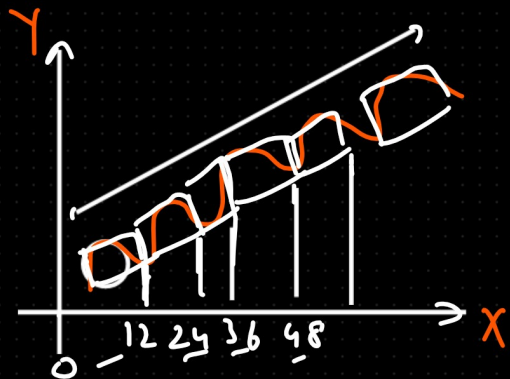
$$\underbrace{(\varepsilon_{t-1} \psi_{t-1} + \varepsilon_{t-2} \psi_{t-2} + \varepsilon_{t-3} \psi_{t-3} - - - - \varepsilon_{t-n} \psi_{t-n} + c)}_{\text{MA part}}$$

AR(1) $\hat{p}=1$
 $= I(2) \hat{d}=2$
MA(1) $\hat{q}=1$

$$y_t = (y_{t-1} \psi_{t-1} + c) + (y_t - y_{t-1} - y_{t-2}) + (\varepsilon_{t-1} \psi_{t-1} + c)$$

SARIMA

↳ seasonal



p, d, q
(P, D, Q)

$(p, d, q)(P, D, Q)_S$

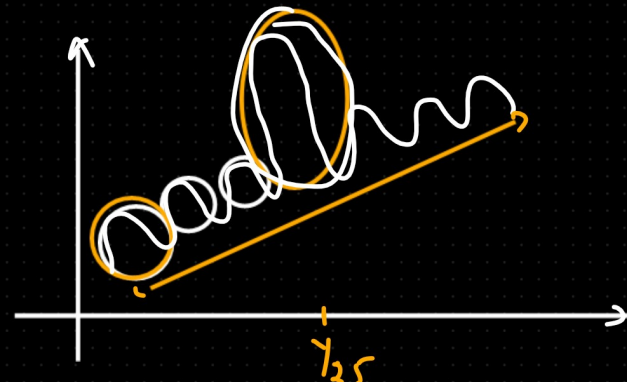
$\Rightarrow (1, 1, 2)(1, 0, 2)_{12}$



SARIMAX

↳ exogenous

(outlier) (intercept)



$(p, d, q)(P, D, Q)_S$

$\Rightarrow (0, 1, 2)(1, 1, 2)_{12}$

